

Smart Withering Monitoring and Control System

Project ID: 25-26J-292

Project Proposal Report

SOORIYAARACHCHI N D

BSc (Hons) in Information Technology Specializing in Software Engineering

Department of Computer Science and Software Engineering

Faculty of Computing

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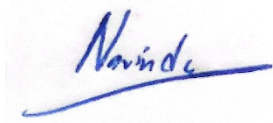
Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION

I declare that this is my own work & this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning & to the best of my knowledge & belief it does not contain any previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Dr. Nathalie De Silva

Signature of the supervisor

Date

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Signature of the co-supervisor

Date

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ABSTRACT

The aim of this research is to develop an intelligent system that helps tea makers regulate the withering process more accurately and conveniently. Withering is a critical first stage in the production of tea, where fresh leaves remove water in preparation for future procedures. Today, this is highly experience and visually reliant on the employees, and this can be a source of error because changes in weather like rain or sun affect how quickly the leaves dry. In order to solve this problem, our system measures two main factors: tea leaves' water content, which is read through a moisture meter, and the temperature around the environment, which affects the way leaves evaporate water. Based on these inputs, the system approximates how much longer it will take the leaves to reach the ideal moisture level of 45%, which is best for high tea quality. In addition, the system takes photographs of the leaves every few hours and identifies them using image recognition software. The system classifies the leaves into one of four withering stages: Raw (75% water content), Early Wilt (65%), Optimal Withering (55%), and Over-Withered (45%). This integrated system tells tea factory staff exactly for how long to continue withering and the condition of the leaves without guessing. Through giving accurate, moment-to-moment feedback, the system avoids problems like over-drying or under-drying of the leaves, saves time, and guarantees the uniformity of the color and taste of the tea irrespective of the erratic weather.

INTRODUCTION

Tea is one of the most widely used beverages globally, and Sri Lanka has long been renowned for producing high-quality tea that is shipped all over the world [2]. Tea making involves a series of processes, and withering has been found to be the initial and most crucial of those processes [3]. During withering, recently harvested leaves lose water in preparation for rolling and fermentation [4]. The accuracy of this stage directly affects the taste, color, and overall quality of the final product [5]. Traditionally, withering has been managed by workers based on experience and sight observation [3]. Although this process is highly subjective and susceptible to being affected by external factors like climate, humidity, and rain, it is found to be difficult to generate consistent results [3], [6].

Previous research on tea processing has focused on the use of sensors, moisture sensors, and image processing to improve quality control [4], [6]. For example, crop water content has been quantified with the use of moisture sensing technology in farming, whereas leaves have been sorted via image recognition techniques according to their physical characteristics. The results show that digital components can be incorporated into the traditional process to reduce human mistakes and improve efficiency [9].

Today, the most developed methods of withering monitoring rely on manual measurement or semi-automatic devices that do not tie together a few parameters such as leaf moisture and environmental conditions in a single solution [7]. While a few scientists tried to execute automation through sensors or computer vision, these solutions tend to be narrow-scope-based and lack real-time feedback for tea factory employees.

Our approach builds on these earlier works by combining sensor technology with image recognition to create an end-to-end integrated smart system. Not only does the system quantitatively measure leaf water content and the ambient temperature, but it also analyzes images of leaves to ascertain their level of withering. This allows for continuous, accurate monitoring and avoids tea farmers from falling into the typical traps of over-withering or under-withering. Through its real-time advice, our method is more precise and practical than the existing literature and guarantees the quality of tea regardless of different weather conditions.

BACKGROUND AND LITERATURE SURVEY

One of the largest farming enterprises in Sri Lanka and one of the leading economic contributors is tea production. Of the various processes of tea manufacture, the initial one and the most crucial one is withering, which is recognized for the reason that it directly affects the physical and chemical nature of the tea leaves. Proper withering allows leaves to shed the appropriate quantity of moisture, making subsequent processes such as rolling and fermentation to take effect[5]. Traditionally, this has been dealt with by skilled personnel who mostly rely on experience, visual judgment, and climatic factors. Nevertheless, this is the most unreliable process because unexpected climatic conditions such as heavy rains or sunshine may slow down or accelerate the rate of withering[8]. Unreliable withering not only affects the final product's flavor, aroma, and color but also decreases the overall quality of tea reaching the final consumers' hands.

With growing technology, there is more room to use digital systems such as sensors and image recognition in tea manufacturing and agriculture. Intelligent systems are now being tested to assist the standardization of production methods, prevent dependency on human instincts, and enhance efficiency. With this background, developing an efficient, automated system to monitor and control the process of withering has been a pragmatic necessity as well as an innovation that can be helpful to the tea industry.

1. Overview of the Domain

Agriculture in recent years has seen a growing reliance on smart technologies such as Internet of Things (IoT), machine learning, and computer vision. Moisture sensors are implemented at scale to monitor water content in crops, and image recognition has been used to grade crop quality, detect diseases, and monitor growth. These technologies demonstrate how digital technologies can be combined with traditional agricultural practices to reduce the scope for human error and enhance efficiency.

2. Overall background of the problem

The fundamental problem in tea withering is the employment of human experience. Leaf workers visually inspect leaves and estimate drying times using experience, leading to inconsistency. There have been studies on the use of moisture meters or environmental sensors within tea factories, but more often than not, such studies focus on a single factor, i.e., leaf water content, without considering the influence of temperature or humidity. Similarly, although computer vision techniques are used in agriculture, their application in tea withering is not adequately studied.

3. Contribution to the Existing Knowledge

This research aims to bridge the gap by bringing multiple methods together in one smart system. The use of measurement of moisture, sensing of the environment, and image recognition in the system offers a more comprehensive approach than what currently exists in stand-alone methods. This is an introduction into the world of smart agriculture because it proves that hybrid sensor-AI systems can be applied to tea production, a sector that has not been as innovated as other crops.

4. Why is this research important

Implications of this research are significant on several fronts:

For the industry

Tea factories will have a systematic way of monitoring the withering process without using mere human intuition. This will eradicate mistakes such as over-drying and under-drying, which often lead to substandard tea quality. Factories will be able to work faster, save time, reduce waste, and generate the same quality in each batch through this system. This makes their process more professional and competitive worldwide.

For fund providers and investors

Investors can have a chance with this research to invest in innovation in an old industry. By funding such an operation, they help tea factories save wasteful costs, improve efficiency, and drive up

profit margins. It also shows how the industry is being revamped with advanced technology, a factor that will attract more purchasers and make Sri Lanka a leading tea-exporting nation.

For society

The consumers who consume tea will enjoy a better product with improved taste, odor, and appearance. The farmers and the laborers will also stand to gain as consistency in quality means higher value for their tea. At a larger scale, this kind of digitalization helps agriculture become smarter and more sustainable, and it can inspire other industries to do the same and invest in technology.

For scientific development

This piece of work benefits the fields of precision agriculture and intelligent systems. This piece of work demonstrates how sensors and image recognition can be employed for solving a real-life problem like tea withering. It can inspire other researchers to employ similar techniques for other crops and operations, developing more innovation in agriculture and food science.

Research Gap

Table 1 – Research Gap table

Feature / Objective	Existing Systems (Research)	Proposed Application Component
Moisture Monitoring	Some studies use moisture meters to measure leaf water content but provide intermittent readings only.	Continuous measurement of tea leaf water content using sensors to calculate progress in drying.
Environmental Monitoring	Temperature or humidity is monitored by small systems, but typically separately and not integrated.	Real-time monitoring of environmental temperature to predict rate of withering.
Image-Based Classification	Computer vision is used in agriculture for disease detection and crop grading but rarely applied to tea withering.	Image recognition to classify tea leaves into withering stages: Raw, Early Wilt, Optimal, Over-Withered.
Integration of Data Sources	Existing approaches focus on single parameters (moisture or temperature or image).	Combines moisture, temperature, and image analysis into one integrated intelligent system.
Real-Time Feedback	Most systems provide offline analysis or require human interpretation.	Automated, real-time feedback and time estimation for optimal withering.
Support for Workers	Relies heavily on worker experience and visual judgment.	Provides clear, objective guidance to staff, reducing dependency on guesswork.
Impact on Tea Quality	Inconsistent results due to manual methods and unpredictable weather.	Ensures consistent quality, saves time, and reduces losses by preventing under or over withering.

Although tea is among the most important export crops in Sri Lanka and in most countries, there has not been much research on the use of new technologies to improve the withering process. Most of the research on tea production has focused on either traditional approaches or single-parameter monitoring, i.e., either moisture only or temperature only. Though these methods help to some

extent, they still fail to provide a complete picture of the withering stage and still require subjective decisions from laborers.

Some researchers have integrated sensor-based monitoring or computer vision methods in agriculture, but there have been very few attempts to consolidate these methods into one system for tea withering. Current solutions also fail to give real-time feedback to factory operators, which is necessary to avoid tea leaves from becoming over-withered or under-withered. Tea producers thus still face the challenge of ensuring consistent quality, reducing errors, and improving efficiency.

This research bridges this gap by designing an intelligent system that integrates moisture measurement, environment monitoring, and image recognition. The system not only forecasts the duration required for the leaves to achieve the ideal moisture content but also classifies the leaves into corresponding withering stages through image analysis. By doing so, this research strives to build a sound and practical tool that can aid tea producers in making accurate, data-supported decisions in the withering process.

In that way, this study assists in filling the gap in the current literature a fully automated real-time solution in tea withering that reduces the dependence on human judgment and ensures consistent tea quality.

Research Problem

Withering of tea is one of the most crucial starting processes in tea production that directly affects the flavor, color, aroma, and quality of tea. Traditionally, tea factory workers use experience judgment and visual observation to determine the correct time when leaves have withered to the ideal level of moisture. It is a highly subjective process and is prone to errors, especially when there are variations in environmental conditions such as temperature, humidity, and sunlight.

The main problems this project aims to solve are:

Inaccurate measurement: Current methods don't provide precise, continuous measurement of either environmental conditions or leaf moisture, which can lead to under-withered or over-withered leaves.

Dependence on human judgment: The process is extremely reliant on worker experience, making it non-uniform and difficult to standardize from factory to factory or batch to batch.

Limited integration of technology: Most existing studies and systems are designed to track one parameter (e.g., temperature or moisture) and rarely combine sensor information with image recognition to assess leaf conditions in real time.

Impact on tea quality and efficiency: The limitations above result in inconsistent tea quality, potential loss of income, and inefficiency in factory processes.

OBJECTIVES

Main Objective

The primary intention of this study is to formulate and design an intelligent monitoring system that can accurately regulate the withering process of tea through sensor-based measurement and image identification. Contrary to conventional withering processes, which heavily depend on the workers' skill and naked-eye observation, the new system will measure tea leaf moisture content and ambient conditions continuously and determine leaf stages from image analysis. By combining these inputs, the system aims to provide real-time prediction and guidance to ensure that tea leaves reach the ideal withering level required for quality tea. Lastly, the dream of this research is to minimize human error, facilitate standardization of withering, and maximize efficiency and uniformity in tea manufacturing.

Specific Objectives

To develop a sensor-based module for tea leaf moisture and environmental temperature continuous monitoring.

Traditional withering relies on manual moisture measurement, not typically precise. With the integration of automated sensors for temperature and moisture, the system shall be capable of delivering accurate, real-time information to enable informed decision-making.

To develop an image recognition model that can detect the tea leaves in different withering stages.

Manual visual examination is extremely subjective. This research will educate and implement a machine learning model to classify tea leaves into stages such as Raw, Early Wilt, Optimal, and Over-Withered, enabling objective evaluation of leaf condition.

To integrate sensor data and image recognition results into a single intelligent system that predicts withering progression.

Existing approaches prefer to use a single parameter in isolation. The proposed system will combine numerous sources of information to provide accurate predictions for remaining withering time, giving a complete solution.

To provide real-time decision support and feedback to tea factory staff.

Employees will be continuously informed and alerted, less dependent on experience-based decisions and less susceptible to error because of imprecise environmental conditions.

In order to examine the system's performance in terms of accuracy, usability, and impact on tea quality.

The final system will be deployed on real tea factory floors to measure how well it improves accuracy in withering, boosts worker productivity, and facilitates consistent high-quality tea delivery.

METHODOLOGY

System Diagram

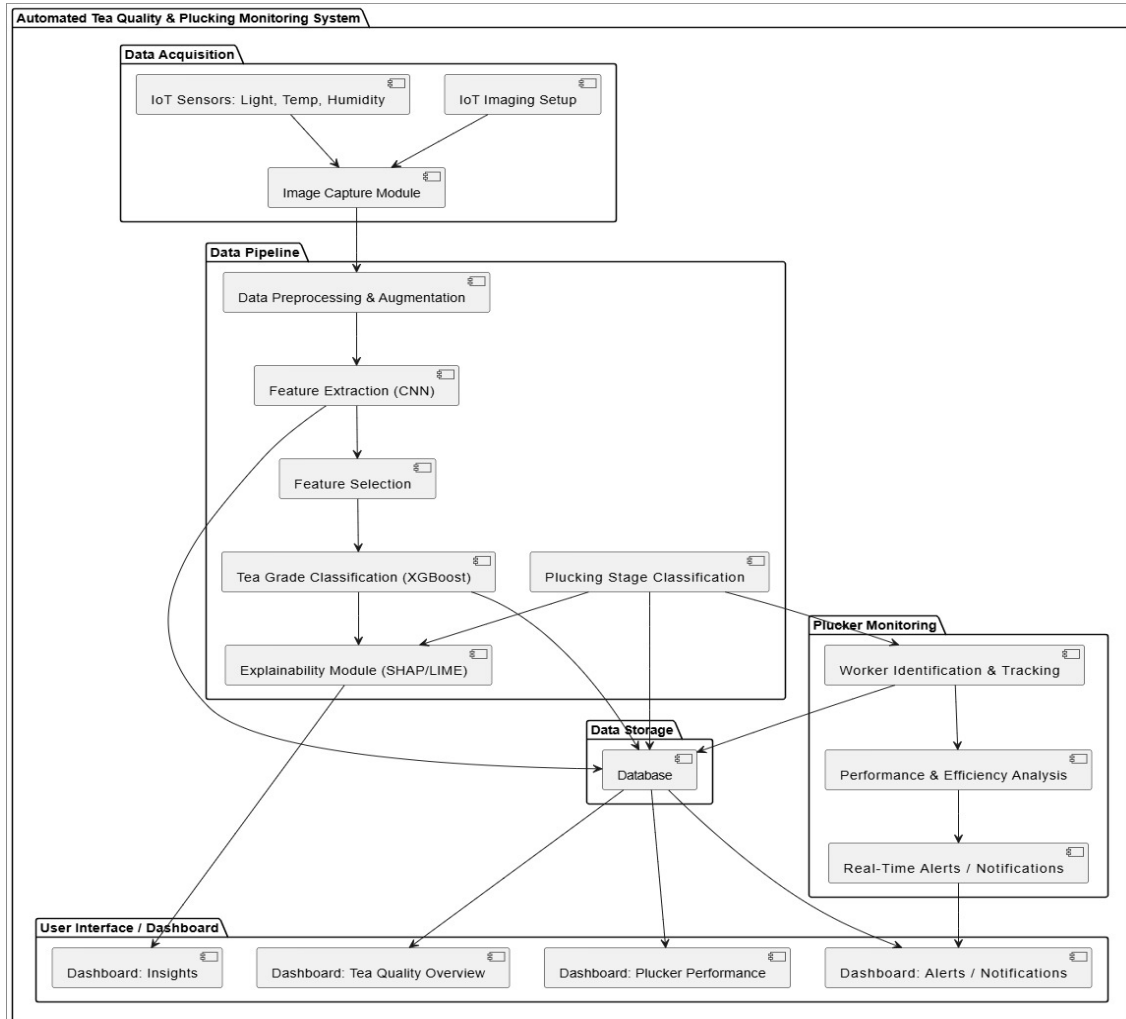


Figure 1- System Diagram

The methodology of the present study intends to integrate sensor-based monitoring and smart image recognition to make a reliable, real-time decision support system for managing the process of tea withering. Unlike conventional methods relying on experience of workers and visual estimation, it utilizes precise sensor measurements (temperature, moisture) and AI-supported image processing to identify withering stages and predict time to reach the optimal condition. The methodology explains the step by step process from requirements gathering to deployment of the system so that the solution becomes deployable in tea factories as well as scientifically acceptable.

This is achieved by the use of environmental and leaf moisture IoT sensors for data collection, machine learning to classify the withering stage, and a user interface for real-time feedback to staff. Robustness, accuracy, and usability for industrial adoption are facilitated by phased structures requirement analysis, system design, prototype development, testing, and field verification.

Application of Key Pillars in the Specialized Area of Knowledge

This study utilizes IoT-based sensing principles, computer vision, machine learning, and precision agriculture:

- **IoT and Sensor Technology:** Leaf water content and environmental conditions will be measured by moisture meters and environmental sensors. These measurements serve as the foundation for accurate predictions.
- **Computer Vision:** Image processing will evaluate ordinary images of tea leaves to classify them based on withering stages (Raw, Early Wilt, Optimal Withering, Over-Withered).
- **Machine Learning:** The classification algorithms (e.g., decision tree models or CNNs) will be trained to recognize withering stages from images and merge predictions with sensor inputs.
- **Precision Agriculture:** Through the integration of data-driven analysis with automation, consistency and efficiency in tea production are ensured.

Application of Technologies in the Relevant Area

The system will apply cutting-edge hardware and software to achieve its objectives:

- **Sensing Hardware:** Digital moisture meters of leaf water content, temperature and humidity sensors DHT22/DS18B20.
- **Image Recognition Frameworks:** OpenCV, PyTorch, or TensorFlow will be utilized to classify and train withering stage images.

- **Prediction Algorithms:** Regression will forecast time to reach 45% optimal moisture content utilizing existing sensor values.
- **User Interface:** Real-time predictions and warnings will be shown on a mobile/web dashboard (React.js, Node.js, MongoDB) to tea factory staff.
- **Deployment:** Configuration based on Raspberry Pi or microcontroller will be responsible for data capture and local processing, with the help of the cloud for storage and analytics.

How to Carry Out the Project

The project will follow an iterative software development lifecycle:

1. Requirement Analysis and System Design

- Define technical specifications of sensors, image-capture devices, and algorithms.
- Design system architecture and data flow.

2. Prototype Development

- Implement first hardware configuration with sensor and camera integration.
- Implement simple dashboard to display sensor readings.

3. Model Training and Optimization

- Collect datasets of tea leaves in different withering stages.
- Train machine learning models to predict stages.
- Tune prediction algorithms for time-to-optimal withering.

4. System Integration

- Integrate sensor data, machine learning models, and user interface.
- Implement alert/notification features for personnel.

5. Testing and Validation

- Conduct lab tests using controlled leaf samples.
- Test in field trials using real tea factories.
- Verify against expert judgments and real-world outcomes.

6. Evaluation and Refinement

- Measure system accuracy, usability, and effectiveness.
- Iterate based on user feedback and error analysis.

Tasks and Sub-Tasks Identified to Achieve Objectives

1. Sensor Module Development

- Select appropriate sensors (temperature, moisture).
- Wiring of sensors with microcontroller.
- Calibration and accuracy testing.

2. Image Recognition Development

- Capture and pre-process images of tea leaves.
- Train machine learning model for stage prediction.
- Testing for robustness under varying light conditions.

3. Prediction Model Development

- Implement regression/ML algorithms for time prediction.
- Integration of predictions with real-time information.

4. System Integration

- Develop dashboard for visualization.
- Integration of sensor + image analysis modules with backend.

5. System Testing and Evaluation

- Lab testing under controlled conditions with known leaf samples.
- Operational tea factory trials in the field.
- Collect feedback and adjust system.

Data Requirements and Collection Methods

- **Sensor Data:** Moisture level (%) and environmental temperature/humidity will be collected continuously.
- **Image Data:** With every few hours' withering, leaf images will be captured.
- **Datasets:** Experimental datasets (created during trials) and literature-supported references for optimal withering moisture level will be used.
- **Preprocessing:** Images will be resized, color normalized, and feature extracted for training models.
- **Validation:** Accounts will be cross-checked with expert evaluations by tea experts.

Commercialization aspects of the project

The intelligent tea withering monitoring system has significant commercial potential for the tea industry in Sri Lanka and internationally. It offers real-time, precise monitoring of leaf moisture and withering stages, reducing reliance on human judgment and prevents both over-drying and under-drying, ensuring consistent tea quality. This reliability can enhance factory productivity, reduce waste, and increase overall profitability, making it an attractive investment for tea producers. The system can be offered as a hardware and software package for small, medium, and large tea factories, with including options for subscription-based software updates, cloud data storage, and ongoing maintenance services. Furthermore, the image recognition and sensor-based monitoring technology can be adapted for other crops or agricultural processes, expanding its market potential beyond tea. By integrating digital technology into traditional tea processing, this system emerges itself as a modern, scalable, and cost-effective solution for improving operational

efficiency and product quality, making it commercially viable for tea manufacturers and agro-tech service providers. Therefore, this can

- Can be marketed as a complete hardware-software solution for small, medium, and large-scale tea factories.
- Offers potential subscription-based software updates, cloud storage, and maintenance services.
- Technology can be adapted for other crops or agricultural processes, expanding market potential
- Cost-effective solution with potential for high adoption by tea producers and agro-tech service providers.

Anticipated Conclusion

The proposed system introduces a functional smart monitoring platform for tea withering that enables accurate, real-time prediction of withering time and effective classification of leaf stages. Reducing the impact of weather changes and removing human bias, the system reduces errors that often arise with manual assessment. This ensures consistent tea quality while enhancing productivity, ultimately supporting the digitalization of agricultural practices within the tea industry.

Gantt Chart

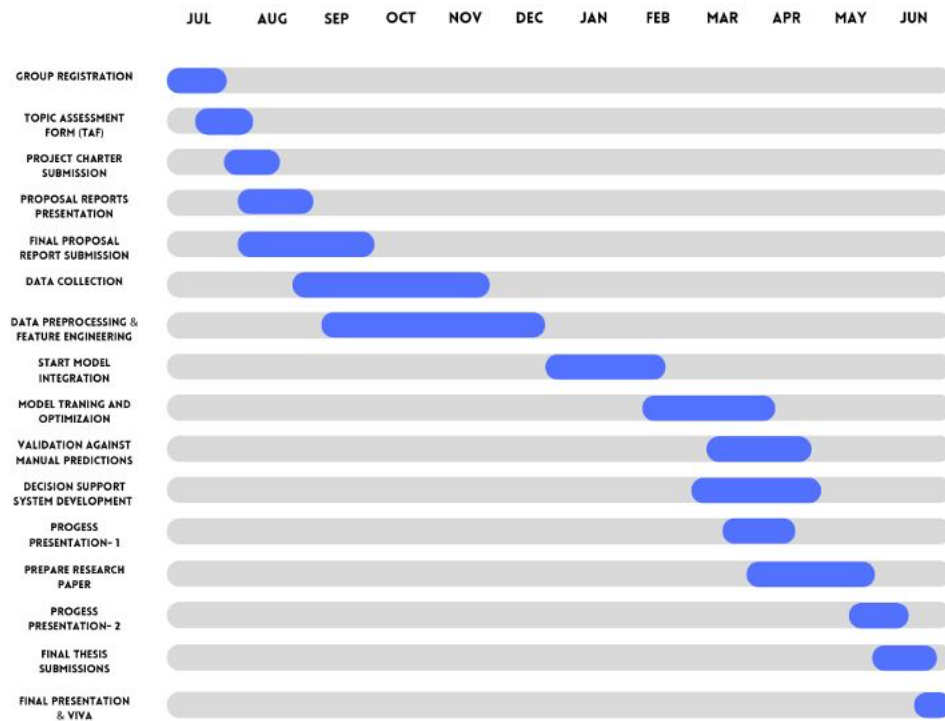


Figure 2: Gantt Chart

PERSONNEL AND FACILITIES

Personnel

Third-year Information Technology undergraduate students with practical experience in IoT, machine learning, and computer vision and the ability to design integrated hardware and software solutions for real-world applications. The team has already gained experience in designing sensor-based systems, environmental data acquisition, and image recognition models with hands-on experience in data processing, analysis, and interpretation. Infrastructure ready for utilization in this project includes temperature and humidity sensors, moisture level meters, microcontroller sets, and Python, OpenCV, and TensorFlow-based test and model developing workstations. Collaboration with nearby tea plantations provides access to new leaf samples, on-site experience, and consultancy from industry experts, and cloud platforms and encrypted data storage ensure data integrity, real-time processing, and scalability. The system is validated through controlled experiments in accurate measurement of tea leaf moisture content and withering stages to enable model verification and assured performance. Periodic documentation of progress, test findings, and model changes allows systematic progress and reporting during the project.

Facilities

The study benefits from several key facilities, including access to the Sri Lanka Tea Board's databases managed by the Information Technology and Statistical Division, which provide extensive past auction data along with current market insights. Data obtained through CICRA Solutions' digital auction platform further supports the analysis by recording bidding behaviors, quality evaluations, and pricing patterns since April 2020. Collaborative agreements with tea industry stakeholders ensure availability of sector-specific knowledge, quality-checking frameworks, and standard datasets are essential for model development. Advanced performance monitoring and analytics platforms are also available, enabling predictive model evaluation, calculation accuracy checks, scalability testing, the use of testing protocols like A/B testing for comparing different machine learning and ensemble approaches.

PROJECT REQUIREMENTS

1. Functional Requirements

Functional requirements describe what the system should do.

- Monitor tea leaf moisture continuously using sensors.
- Monitor the environmental temperature and humidity around the leaves.
- Capture intermittent photographs of tea leaves for analysis.
- Classify leaves into four withering stages: Raw, Early Wilt, Optimal Withering, Over-Withered.
- Predict the time left for leaves to reach optimal moisture (45%).
- Provide real-time notifications or alerts to staff on leaf status.
- Generate reports on withering progress for factory management.
- Store data for future analysis and decision-making.

2. Non-Functional Requirements

Non-functional requirements specify how good the system is.

- Accuracy: Temperature and moisture readings should be precise in order to make precise predictions.
- Reliability: The system should run faultlessly and continuously during production hours.
- Usability: The interface should be user-friendly for factory workers with minimal training.
- Performance: The system should process sensor data and image analysis in real time.
- Scalability: The system should be capable of managing multiple trays or batches simultaneously.
- Security: The information should be kept in a safe form and should be accessed by authorized individuals only.

- Maintainability: The system should be simple to update and maintain.

3. User Requirements

- User requirements specify what the system needs to provide for the end users.
- The factory staff should be capable of viewing leaf moisture and stage in real time.
- Staff should be notified when leaves reach optimal or critical stages.
- Management should be able to generate and export reports on withering processes.
- Users should be able to view historical data to analyze trends.
- Users should require minimal training to operate the system.

4. System Requirements

Hardware:

- Moisture sensors for measuring leaf water content.
- Environmental temperature and humidity sensors for monitoring.
- Digital camera for capturing leaf images.
- Microcontroller or processing unit to merge sensor inputs and control the system.
- Display or interface device (PC/tablet) for interaction by personnel.
- Storage device for saving sensor and image data.

Software:

- Image recognition software for classifying leaf stages.
- Software for analyzing sensor readings and estimating withering time remaining.
- User interface application for real-time monitoring and alerts.
- Database for logging historical measurements and reports.

BUDGET AND BUDGET JUSTIFICATION

The overall budget of the entire project Rs. 27,500.00 is a smartly planned and economically sensible investment on four critical components, which cannot be avoided in the process of creating a complete tea auction price prediction system. The fact that Rs. 10,000.00 (36.4%) will be spent on software licenses and tools is also central to providing a secure technical implementation such as scalable deployment of ML models, professional development environments and special APIs. The travel costs of Rs. 7,500.00 (27.3%) support the necessary field research, like visiting tea gardens in different parts of it, viewing events in auction centers and even meeting with stakeholders thereby offering invaluable ground-truth information and empirical confirmation. The allocation of Rs. 5,000.00 (18.2%) to meteorological and external data will ensure access to important weather patterns, global market indices and economic indicator to ensure that accuracy is achieved to the model beyond the local auction parameters. The other Rs. 5,000.00 (18.2%) represent the research materials needed to academic resources, research papers, industry reports, and specialized publications on tea market analysis necessary for comprehensive analysis and professional presentation of results. The balance of the budget substantially maximizes research impact in its use of open-source technologies strategically and yet having enough resources in data quality, technical infrastructure, and actual validation offering unique value to the research community of the Sri Lankan tea industry.

Table 2: Expected Budget

Requirements	Cost (Rs)
Software Licenses and Tools	10000.00
Travelling Cost	7500.00
Meteorological and External Data	5000.00
Research Materials and Resources	5000.00
Total	27500.00

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